

Global Population Data Scraper

Background

Organizations such as research firms, policy institutes, and data science teams frequently rely on publicly available datasets to analyze global trends. Wikipedia maintains regularly updated statistical tables that serve as reliable sources of structured public data. As a Data Engineer, your task is to build an automated system that collects country-level population data from a live webpage and converts it into a structured dataset for analysis.

Objective

Develop a Python-based web scraper that extracts population data of countries from Wikipedia and stores it in a clean, structured format suitable for further analysis.

Data Source

https://en.wikipedia.org/wiki/List_of_countries_and_dependencies_by_population

Functional Requirements

- Connect to the webpage by sending an HTTP request and verify the response using the status code.
- Identify the correct population table through HTML inspection.
- Extract the following fields: Country/Territory, Population, Date, Source.
- Store the extracted data in a file named 'countries_population.csv' with appropriate column headers.

Expected Outcome

You should produce a working Python scraper, generate a structured dataset, and write code that is readable and logically organized. Treat this task as building a miniature data pipeline.